

Lemma 3.1. For $f \geq 0$ on G^t , the product of f over the 2^t vertices of a random box has mean at least $(\mathbb{E}f)^{2^t}$; so a random box in $\{0, \dots, L\}^t$ is monochromatic with probability at least k^{1-2^t} .

run along t chains of L points; discard degenerate boxes

Proposition 3.2. For every $r \leq n - M$ some shape $(u; a_1, \dots, a_t)$ with $r \leq u \leq r + M - t$ satisfies $\rho_\chi(u; a) \geq k^{-2^t}$.

tune $t + 1$ block sizes, fix the rest

Lemma 4.1. For each level j and generator set A , at least M^{m-t-1} of the M^m size vectors β give the cube $Q(j, A)$ a good shape.

average over (σ, β)

Lemma 4.2. Some ordering σ and sizes β make at least $|\mathcal{P}|/(k^{2^t} M^{t+1})$ of the cubes monochromatic at once.

union rule, Lemma 2.4

Proposition 2.5. Monochromatic cubes of one block chain glue into a monochromatic union-closed family, so
 $F_k(n) \geq |\mathcal{P}|/(k^{2^{t+1}} M^{t+1}) \geq c_{k,t} n^{t+1}$.